

Colorings with Fractional Defect
or
How to 2-color a Graph

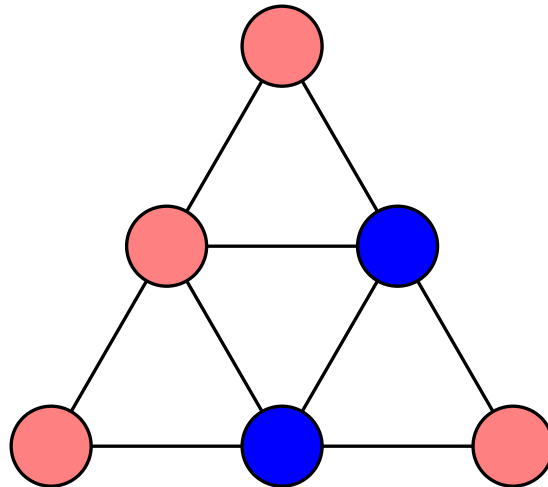
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An Example: Defect

What's the best 2-coloring of a graph?

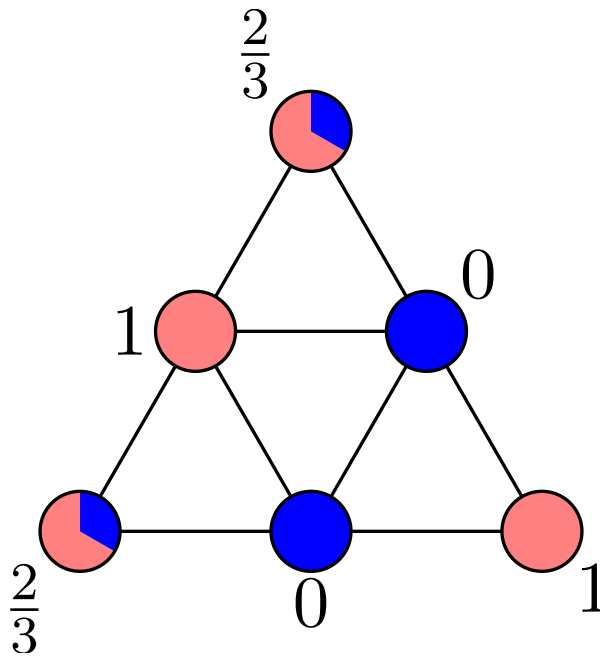
One measure of optimality is defect.

Best defect we can get for 3-sun is 2:



An Example: Fractional Coloring

But if we allow fractional color as follows, the best coloring has defect $\frac{4}{3}$.



Definitions

We color each vertex v_i with fractional r_i *Red* and b_i *Blue*. Such that $r_i + b_i = 1$.

Defect of vertex v_i is sum of overlaps over all colors and all neighbors of v_i :

$$defect(v_i) = \sum_{v_j \sim v_i} \min(r_i, r_j) + \min(b_i, b_j)$$

The Parameter

Goal: for given number of colors (in this talk always 2), minimize the *maximum* defect over all vertices v_i .

Call this $D_2(G)$.

Exists: (note continuity, convexity, linear program)

Triangle

Color one vertex *Red*, one vertex *Blue*, and other vertex arbitrarily.

Fact. We may always assume one pure *Red* node and one pure *Blue* node.

Aside

If one wants to minimize the *sum* of the defects, then that can be achieved by every vertex pure.

Odd Cycle

If we just did ordinary coloring, must have defect 1. This is still best possible.

Fact. If $D_2(G) > 0$ then $D_2(G) \geq 1$.

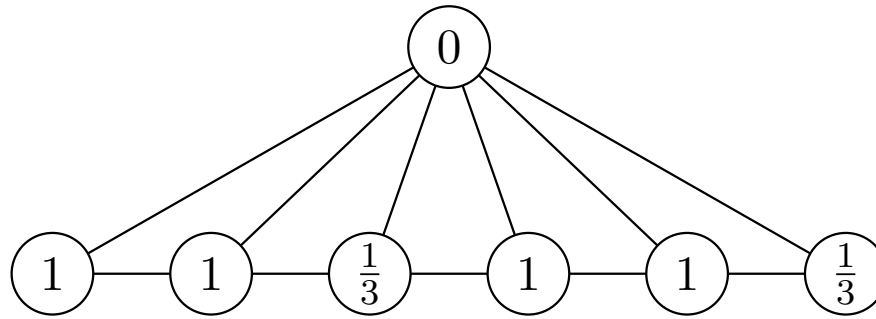
Complexity

It is NP-hard to determine if there is a coloring with maximum defect at most some specified d .

(Proof: If $d = 1$ achieved, can be achieved by pure vertices; then use known complexity of defective coloring (Cowen).)

Fans

Here is the fan F_6 :



Theorem. $D_2(F_n) = \frac{2\lfloor n/3 \rfloor}{\lfloor n/3 \rfloor + 1}.$

Outerplanar Graphs

The fan is outerplanar.

Known that one can ordinarily 2-color an outerplanar graph with defect at most 2.

Best possible? (Certainly almost.)

Complete Multipartite

Let K_a^m be complete m -partite graph with a vertices in each set.

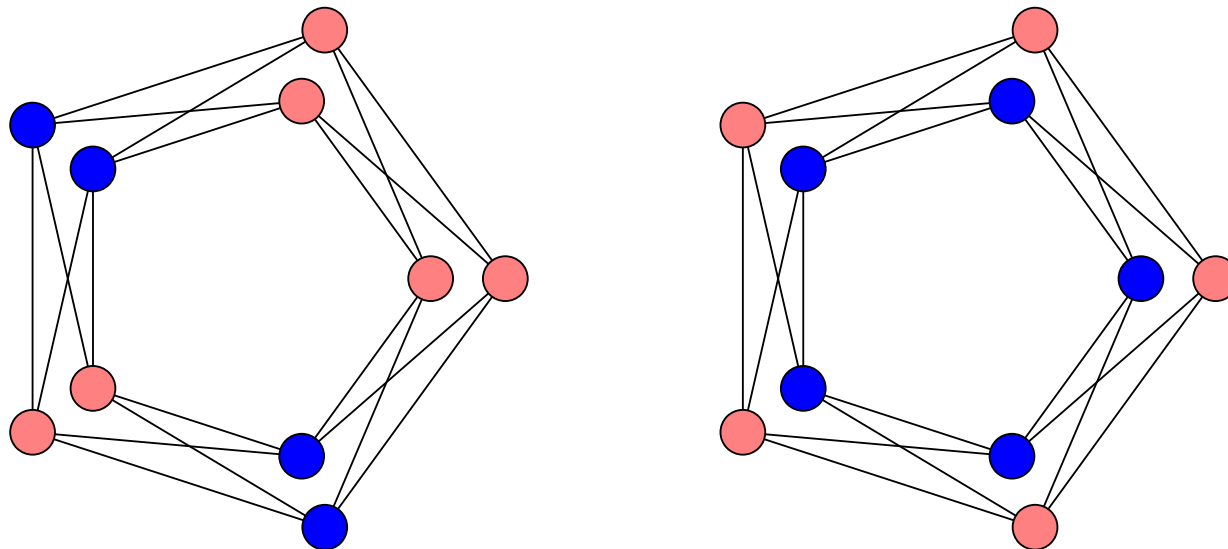
Theorem. $D_2(K_a^m) = (\lceil m/2 \rceil - 1)a$.

Construction: all vertices pure, half partite sets *Red*, half *Blue*

Best possible: takes some work

Expanded Odd Cycle

Theorem. For m odd, $D_2(C_m[2K_1]) = 2$.



Rooks Graphs

Let $R(m, n)$ be $m \times n$ Rooks graph (Cartesian product of K_m and K_n).

Result is easy if m and n both even: $(m+n)/2 - 2$.
(Color each row and column, half *Red* and half *Blue*.)

Result is not easy in general.

Rooks Graph

Here is coloring showing $D_2(R(3, 5)) \leq 38/13$.

Red-usage:

0	8/13	0
0	0	8/13
1	11/13	0
1	0	11/13
6/13	1	1

4-Regular Graphs

Lovász showed that $D_2(G) \leq 2$.

Computer search suggests:

Conjecture. Apart from expansion of cycle $G = C_m[2K_1]$ where m is odd, it holds that $D_2(G) < 2$ for all but finitely many connected 4-regular G .

And Now For Something Completely Different

Supper.